

Solutions Reference

1. Functions & Overloading

```
double calculateOctave(double freq) { return freq * 1.414; }

void logSignal(double val) { /* print val */ }
void logSignal(double val, int band) { /* print band and val */ }
```

3. Lambdas & Signal Logic

```
// Subtracting max PSD using Lambda
auto subtractor = [](double current, double maxPsd) {
    return current - maxPsd;
};

void process(double signal, auto func) {
    double result = func(signal, 100.0); // 100.0 is max PSD
}
```

6. Fold Expressions (Variadic)

```
template<typename... Args>
auto averageSignal(Args... args) {
    double sum = (... + args);
    return sum / sizeof...(args);
}
```

7. Constexpr & Type Constraints

```
template<typename T>
requires std::is_arithmetic_v<T>
constexpr T computeStaticGain(T input) {
    return input * 0.5;
}
```